

Confidential Technical Feasibility Brief

HSLA-like Steel Architecture for Coupled Corrosion-Fatigue and Crack-Arrest Screening

Safe External Version v0.2 | For laboratory feasibility discussion only

Confidential discussion draft: this document intentionally omits the underlying reasoning framework, internal scoring method, optimization logic, formula structure, weighting design, and patent-claim draft. Its only purpose is to let a metallurgy laboratory assess whether a feasible small-sample route exists.

v0.2 update: adds a dual-baseline B0 control logic and explicitly notes the cost, schedule, and statistical limits of coupon-level G2 corrosion-fatigue and G3 crack-arrest testing. This reduces the risk of false positives from weak baselines or oversimplified tests.

1. Purpose

This brief asks whether a metallurgy laboratory can evaluate, at coupon-level feasibility, an HSLA-like steel architecture targeted at marine structural failure modes. This stage is not a request for a complete alloy formula or production-grade steel design. It converts a failure-chain hypothesis into a safe, testable laboratory problem.

Central question: can a weldable HSLA-like matrix support both corrosion-fatigue-control behavior and crack-arrest behavior without obvious loss of weldability, HAZ stability, or plate-scale manufacturing potential?

2. Scope and Non-Disclosure Boundary

This document is deliberately limited to outer technical requirements. It does not disclose the internal method used to generate them.

Discussable content	Not disclosed in this version
Target failure chain and laboratory test gates	Internal reasoning framework, scoring logic, or formulas
Functional sample-grouping concept	Precise optimization process or weighting method
Desired observable behavior: crack arrest, corrosion-fatigue decoupling, HAZ stability	Alloy composition ranges, heat-treatment recipes, or proprietary design rules
Feasibility questions for the laboratory	Patent-claim draft, military-performance claims, or production steel specification

3. Technical Problem Statement

Marine structural steels often fail through coupled rather than isolated degradation. Early architecture screening should therefore not maximize only yield strength, hardness, or standalone corrosion rate. It should test whether failure transitions are delayed or interrupted.

marine environment -> localized corrosion / pitting -> cyclic fatigue loading -> pit-to-fatigue crack transition -> crack initiation -> crack-path connectivity -> accelerated crack growth -> weld / HAZ amplification -> structural failure risk

The proposed evaluation should focus on three coupled transitions: pit-to-fatigue conversion, crack-path connectivity, and weld / HAZ amplification.

4. Proposed Functional Architecture

The proposed architecture is not a finished alloy. It is a screening direction:

HSLA-like weldable matrix + crack-arrest functional field + corrosion-fatigue-control functional field

The HSLA-like matrix is the engineering anchor. Crack-arrest and corrosion-fatigue-control functions are candidate enhancements, but they must not degrade weldability, HAZ behavior, or scale-up potential.

5. Baseline Control: Dual B0 Logic

The baseline material should not be selected merely because it makes improvement easy to show. A weak B0 can create false improvement; an overly strong B0 can block a legitimate feasibility screen. A dual-baseline logic is recommended.

Baseline	Role	Suggested options	Interpretation limit
B0-A physical test baseline	First-round coupon control; must be obtainable, processable, and weldable	Prefer API 2W Grade 50 or API 2H Grade 50; if difficult, use ASTM A131 / ABS DH36 or EH36	Improvement over B0-A is a feasibility signal only, not proof of superiority over advanced naval steel
B0-B literature / strategic benchmark	High-level reference to prevent over-interpretation	HSLA-80 / HSLA-100-class naval structural steel or another advanced public benchmark	Not required as a first-round physical coupon; used only to constrain interpretation and future targets

Practical guideline: if the laboratory is oriented toward offshore, thick-plate, or welded structures, ask first about API 2W Grade 50. If ship-plate availability is the main constraint, use ABS/A131 EH36 or DH36 as the substitute B0-A.

6. Suggested Coupon-Level Screening Groups

The laboratory may adjust implementation based on equipment, metallurgical route, safety limits, and prior art. The groups below are concepts, not composition instructions.

Group	Concept role	Primary observation	Hard boundary
B0-A	Physical baseline	Reference strength, toughness, corrosion-fatigue, and welding response	Must not be a weak baseline selected to manufacture apparent improvement
B1 CA-oriented	Crack arrest / deflection / blunting	Interrupted crack connectivity or slower crack growth	Weldability must not fall below baseline
B2 CF-oriented	Corrosion-fatigue decoupling	Delayed pit-to-fatigue conversion or lower corrosion-fatigue sensitivity	No embrittlement or HAZ-sensitive behavior
B3 CA-CF combined	Observe crack arrest and corrosion-fatigue control together	Best coupled-risk suppression without single-property collapse	Must retain matrix-scale manufacturability potential
B0-B	Literature / strategic reference	Constrains interpretation and future targets	Not used for first-round victory claims

7. Recommended Early Test Gates

The first laboratory goal should be gate-based feasibility, not final performance optimization.

Gate	Question	Preferred evidence type
G1 Weldability / HAZ stability	Does the candidate avoid obvious welding or thermal-cycle degradation relative to baseline?	Simulated HAZ, welded coupons, hardness / microstructure gradients, basic mechanical screen
G2 Corrosion-fatigue behavior	Does the candidate reduce pit-to-fatigue conversion risk or corrosion-fatigue sensitivity?	Pitting exposure + cyclic loading comparison, crack-initiation statistics, surface / fracture analysis
G3 Crack-arrest behavior	Does the candidate interrupt crack connectivity, deflect cracks, blunt crack tips, or reduce growth rate?	Fractography, crack-growth tracking, microstructure-path correlation
G4 Coupled-risk balance	Does one improvement avoid degrading other critical functions?	Multi-metric comparison to baseline, not single-metric selection
G5 Manufacturability pre-check	Does the direction retain thick-plate, welding, inspection, and later scale-up potential?	Laboratory assessment of process complexity, cost, repeatability, and scale risk

8. G2 / G3 Cost and Statistical Limits

G2 and G3 are the most valuable tests and also the most likely to raise cost and schedule. The first round should not demand lifetime prediction or engineering certification. It should screen for whether a positive signal exists.

Item	Acceptable first-round target	Avoid
G2 corrosion-fatigue	Compare B0-A and B1/B2/B3 under the same exposure and cycling conditions for crack initiation, pit-to-crack conversion, or sensitivity differences	Do not replace corrosion-fatigue judgment with only corrosion rate or hardness
G3 crack arrest	Observe crack deflection, blunting, connectivity interruption, lower growth rate, or fracture-path differences	Do not claim crack arrest from impact value or tensile value alone
Statistical interpretation	Small repeats and microscopy evidence are acceptable in round one; describe results as signals, not conclusions	Do not describe one or two improved coupons as production-level performance

9. Questions for the Laboratory

- Among API 2W Grade 50, API 2H Grade 50, and ABS/A131 DH36/EH36, which is easiest for the laboratory to obtain and use as a credible B0-A control?
- Can HSLA-80/100-class material be treated as B0-B literature reference only, rather than a mandatory first-round physical control?
- Which non-sensitive microstructural or processing mechanisms could plausibly support crack-arrest behavior while preserving weldability?
- Which non-sensitive mechanisms could reduce pit-to-fatigue conversion or weld / HAZ corrosion-fatigue amplification?
- For first-round feasibility rather than certification, what minimum sample count, exposure condition, cyclic loading method, and microscopy package are realistic for G2/G3?
- What obvious failure risks would make this direction unsuitable before any small-sample fabrication?

10. Success Criteria for a First Feasibility Round

A first feasibility round is positive only if the combined CA-CF direction shows at least preliminary coupled-risk improvement while preserving baseline feasibility. If one sample has the highest strength but worse HAZ, corrosion-fatigue, or crack-connectivity behavior, it should not be selected as the winning route.

Positive signal: improved corrosion-fatigue or crack-path behavior relative to B0-A without obvious weld / HAZ degradation.

Neutral signal: no improvement over B0-A but no major degradation; architecture may need redesign.

Negative signal: single-property gain accompanied by weldability, HAZ, corrosion-fatigue, or manufacturability collapse.

11. IP and Collaboration Note

This brief is suitable for confidential technical discussion only. Before exchanging composition ideas, processing recipes, unpublished laboratory know-how, or sample results, the parties should consider an NDA or other written confidentiality arrangement. If the laboratory identifies a concrete route and early data are positive, patent counsel should evaluate protection options: method, screening workflow, material architecture, process conditions, or later concrete material claims.

12. One-Page Summary

This project is not asking for the strongest steel by a single metric. It asks whether a weldable HSLA-like steel architecture can be screened for lower coupled failure-chain risk, especially pit-to-fatigue conversion, crack-path connectivity, and weld / HAZ amplification. v0.2 explicitly requires dual B0 baseline control and written G2/G3 interpretation limits to avoid false positives and over-claims.